POLICY BRIEF · TEACHER LABOR MARKETS · JULY 2026

Who leaves teaching? Teacher attrition in the United States, 2005–2025

Evidence from 174,872 school teachers followed in linked Current Population Survey microdata

EXECUTIVE SUMMARY

- Each year, **15.1%** of degree-holding school teachers leave the profession and do not return within our observation window. The conventional 12-month measure gives 17.6%, but one in six of those leavers is teaching again within three months.
- Persistent attrition rose from about 13% in 2005–2010 to about 17% in 2022–2024, with a peak in the pandemic year and a new high of 18% in 2024.
- Part-time work is the strongest predictor of leaving (+12 percentage points). Risk is U-shaped in age, lowest around 40. Black, non-citizen and preschool teachers leave more. The results are almost identical under both definitions.
- The typical leaver is a woman in her late career, working part-time, without a graduate degree. The highest-risk decile has a 35% exit probability, 2.4 times the average.

15.1 %
of teachers leave the profession for good each year

+12 pp
extra exit risk from part-time work

1 in 6
12-month leavers is back teaching within 3 months

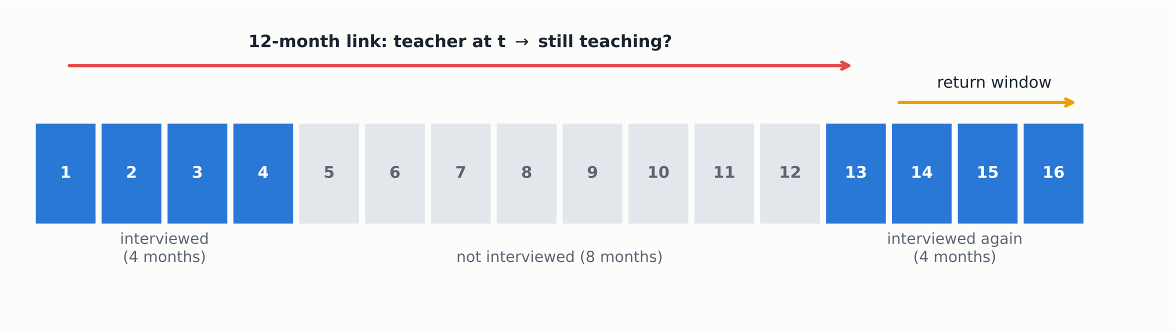
1. Data

We use Current Population Survey (CPS) basic monthly microdata: 251 monthly files from January 2005 to December 2025, obtained from the U.S. Census Bureau (2024–2025) and the NBER mirror (2005–2023). October 2025 was never released.

The CPS interviews each household on a 4-8-4 rotation (Figure 1): four consecutive months in sample, eight months out, four months back in. A person is therefore observed for at most 16 months, and only once. Within that window, each adult interviewed in year t is re-interviewed in the same calendar month of $t+1$. We link the two interviews with household and person identifiers and validate each match on sex, race and age progression. Nobody can be followed across several years, so each base year is a fresh cohort and every rate in this brief is computed wave by wave.

Table 1 shows the population funnel, pooling all 251 months.

Figure 1: The CPS follows each person for at most 16 months. Interview calendar of one rotation group; numbers are months since first interview.



The 12-month link compares months m and $m{+}12$ ($m \le 4$). The return window re-observes leavers in months $m{+}13$ to 16.

Table 1: From the CPS universe to the analytic sample.

	Person-months	% of previous
Adult interviews (person-months), 2005–2025	23,752,044	–
employed	14,450,470	61%
school teachers (occ. 2300–2330)	525,651	4%
with bachelor's degree or higher	464,144	88%
in a linkable rotation (MIS 1–4)	229,930	50%
Linked to their interview 12 months later	174,872	–
with re-interviews after $t{+}12$ (MIS 1–3) → **main sample**	129,897	74%

Teachers are employed persons whose primary-job occupation is preschool/kindergarten, elementary/middle, secondary or special-education teacher (Census codes 2300–2330, identical across the 2002, 2010 and 2020 classifications), with a bachelor's degree or higher. Only rotations in months-in-sample 1–4 can be linked forward. 67% of linkable teachers are found and validated 12 months later; most losses occur because the CPS samples addresses and does not follow people who move.

Defining a leaver. The conventional measure, used by the NCES Teacher Follow-up Survey, counts anyone not employed as a school teacher twelve months after baseline. Temporary exits such as parental leave or a short unemployment spell inflate it: 15.7% of those leavers are teaching again within three months. Our main measure is therefore the persistent leaver: out of teaching at $t{+}12$ and in every later re-interview. It is computed on the 129,897 teachers whose rotation allows at least one re-interview. Exits longer than a year that end in a return cannot be detected, so persistent attrition is still an upper bound on permanent exit, just a much tighter one. All rates use CPS person weights.

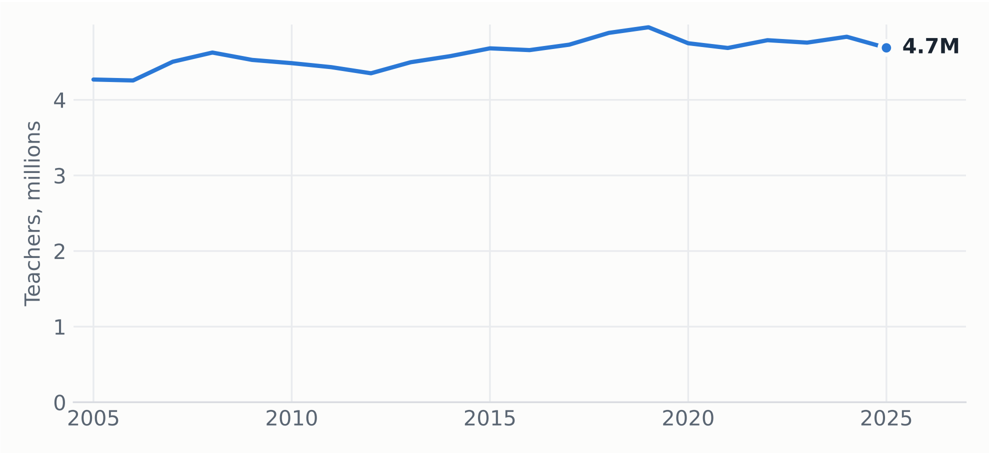
Caveat. Occupation in the returning rotation is re-coded independently, which inflates measured switching. Levels should be read as upper bounds; comparisons across groups and over time, the object of this brief, are far less affected.

2. The teacher workforce

Weighted to population, the CPS counts a stable stock of about 4.7 million degree-holding school teachers (Figure 2), most of them in elementary and middle school (Figure 3).

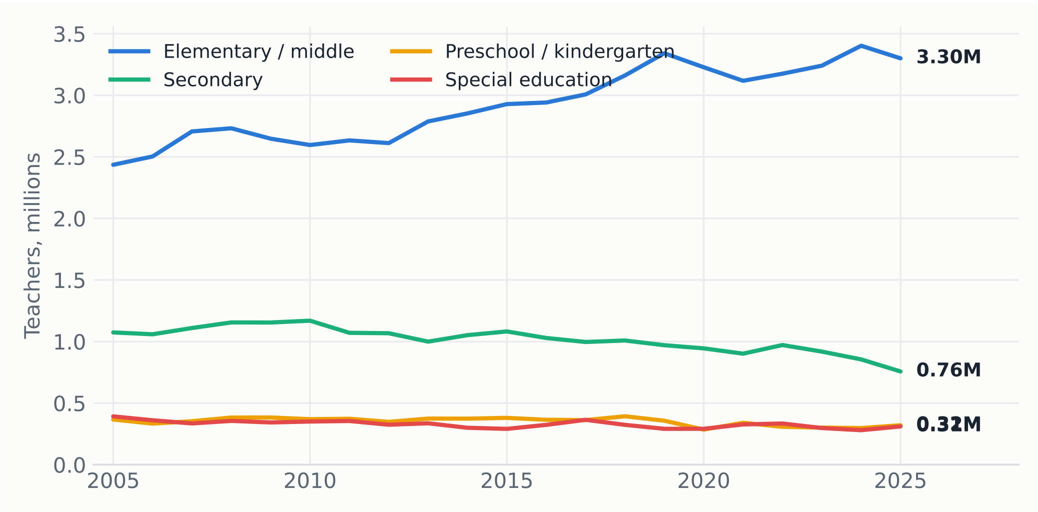
The linked pairs also show both sides of turnover (Figure 4). Teaching loses and recruits 15 to 20% of its stock every year. It is a high-churn occupation, and in recent years exits have run

Figure 2: The stock of school teachers. Employed school teachers with a bachelor's degree or higher, annual average of monthly estimates.



Source: CPS basic monthly files, person weights. Teachers: employed, primary-job occupation codes 2300–2330, bachelor's degree or higher.

Figure 3: Teachers by teaching level.



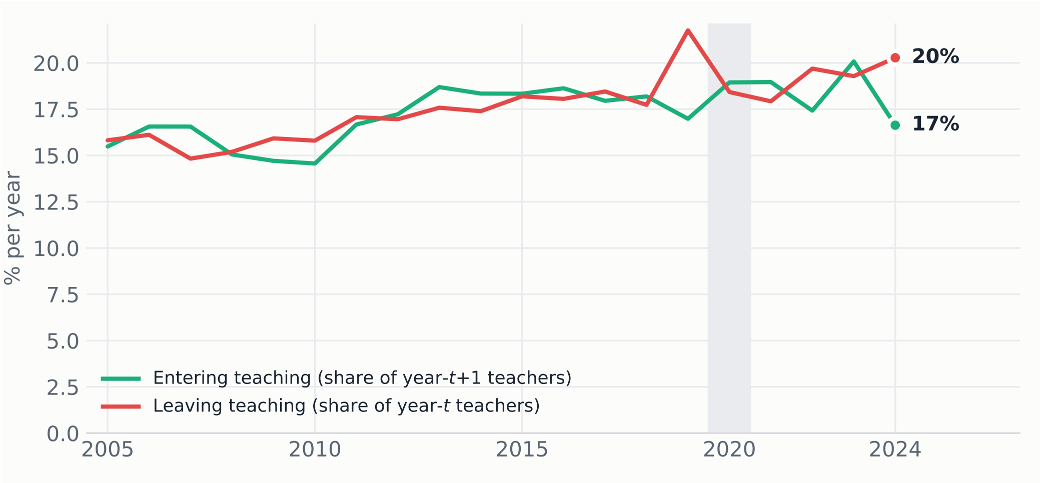
Source: CPS basic monthly files, person weights. Same identification as Figure 2, split by occupation code.

above entries. Most persistent leavers move to another occupation (9.9 of the 15.1 points); 4.5 points leave the labor force and 0.6 become unemployed.

3. Attrition over time

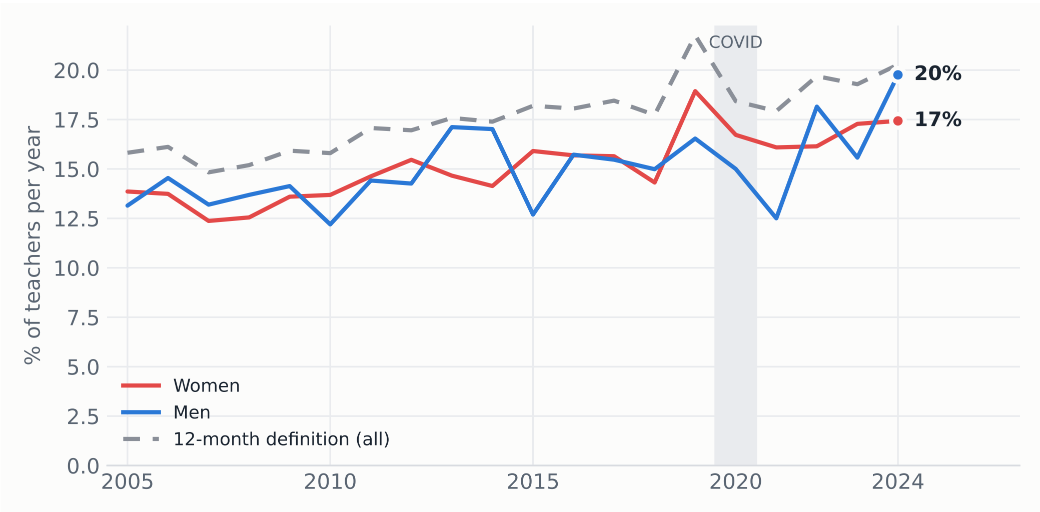
One in six 12-month leavers is teaching again within three months (women 16.1%, men 14.0%). These are temporary exits that the persistent definition strips out. Net of this churn, persistent attrition still climbed from about 13% in 2005–2010 to about 17% in 2022–2024 (Figure 5), peaking in the pandemic year (18.4% in 2019→2020) and reaching a new high of 18.0% in 2024. Gender gaps are small and unstable; in the most recent year men (19.8%) exceeded women (17.4%).

Figure 4: Teachers entering and leaving each year. Gross annual flows in the linked pairs, 12-month definitions.



Source: linked CPS pairs. Entering: school teacher (BA+) at $t{+}12$ who was not a school teacher at t. Leaving: the reverse. Linked samples under-count movers, so both flows are conservative.

Figure 5: Persistent attrition by year and gender.



Source: authors' calculations, linked CPS 2005–2025. The return window is capped at three monthly interviews after $t{+}12$; 21,873 of 30,244 leavers are re-interviewed at least once.

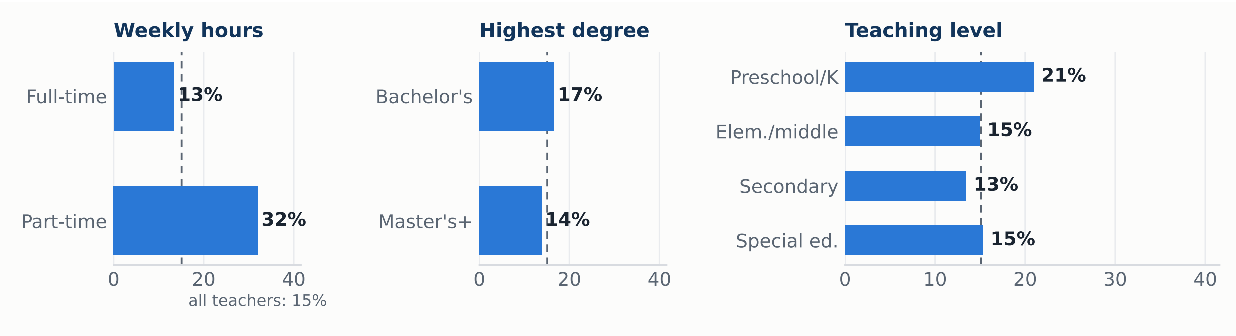
4. Who leaves

Attrition concentrates in three margins: hours, teaching level and, more weakly among degree holders, credentials. Part-time teachers leave at more than twice the rate of full-time teachers, and preschool/kindergarten is the most exposed level (Figure 6).

5. The model

We estimate a probit for the probability of persistently leaving, with covariates measured at baseline, base-year fixed effects and standard errors clustered by household (Table 2). Figure 7 plots the average marginal effects.

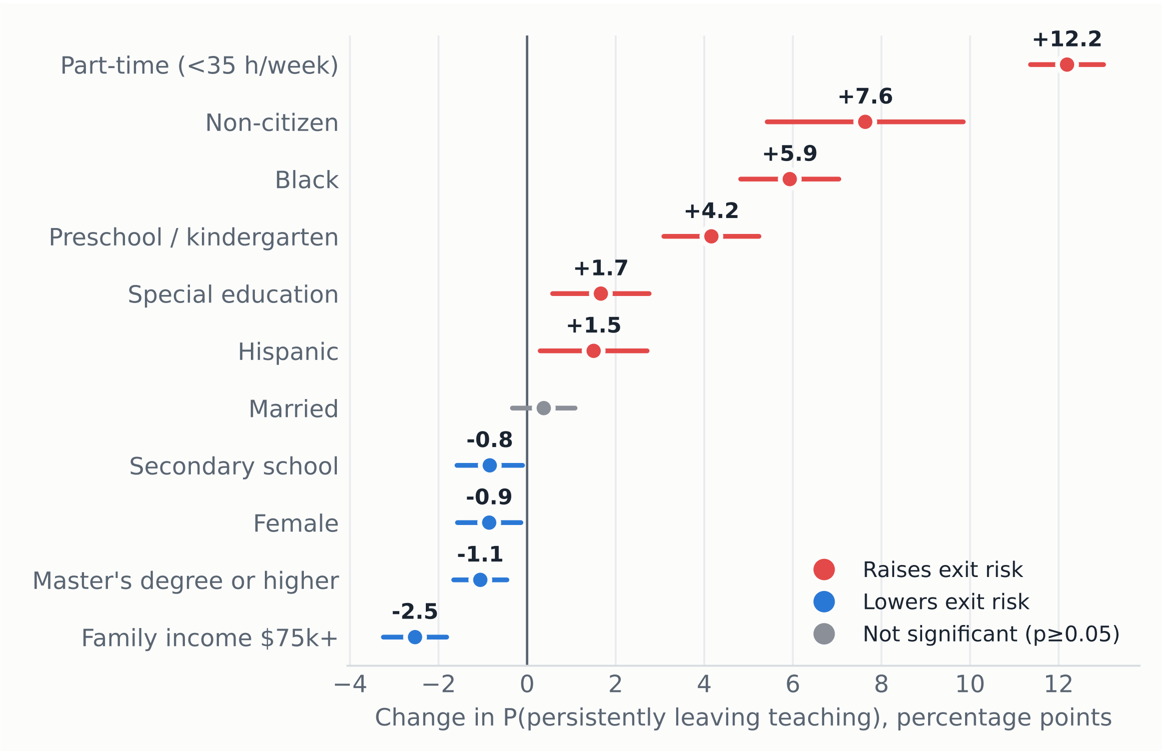
Figure 6: Persistent attrition by baseline characteristics, 2005–2025.



Source: authors' calculations, linked CPS 2005–2025. Dashed line: all-teacher average (15.1%).

Three results stand out. First, part-time status raises exit probability by 12.2 points, the largest effect in the model by an order of magnitude: attachment to full-time hours, not demographics, is the dominant margin. Second, minority and non-citizen teachers are significantly more likely to leave (Black $+5.9$, non-citizen $+7.6$, Hispanic $+1.5$ points), a retention gap that matters for workforce diversity. Third, among degree holders a graduate degree adds only 1.1 points of protection, while higher family income, secondary school and being a woman also retain modestly. Every effect is nearly identical under the conventional 12-month definition (part-time: $+14.5$ vs $+12.2$), so the predictors do not depend on how leaving is measured.

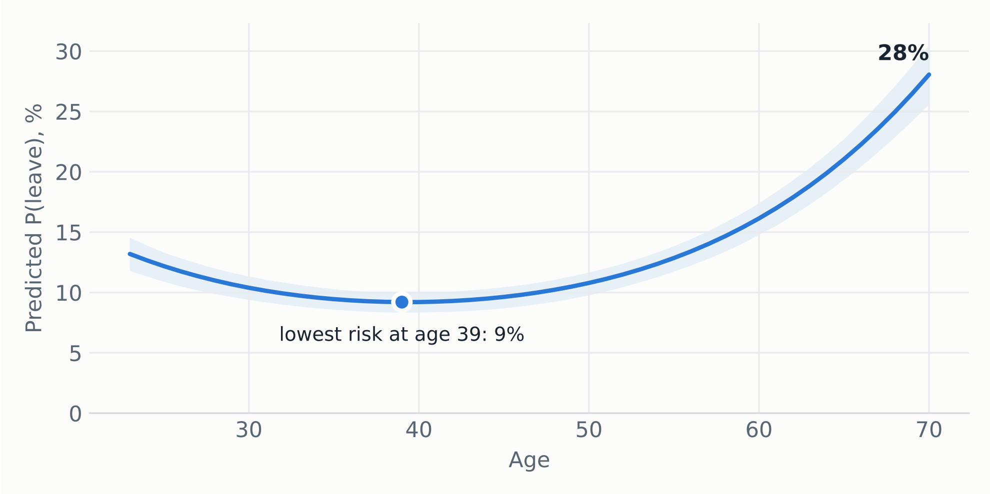
Figure 7: Average marginal effects on the probability of persistently leaving teaching, with 95% confidence intervals.



Source: probit of Table 2. Age terms shown in Figure 8.

Age enters with a robust U-shape (Figure 8). Predicted risk falls to a minimum of 9% around age 39, accelerates after 55 as retirement approaches, and reaches 28% at 70.

Figure 8: Predicted probability of leaving by age, other covariates at sample means. Shaded band: 95% confidence interval.



Source: probit of Table 2, simulation-based confidence band.

Table 2: Probit estimates: probability of persistently leaving teaching, 2005–2025.

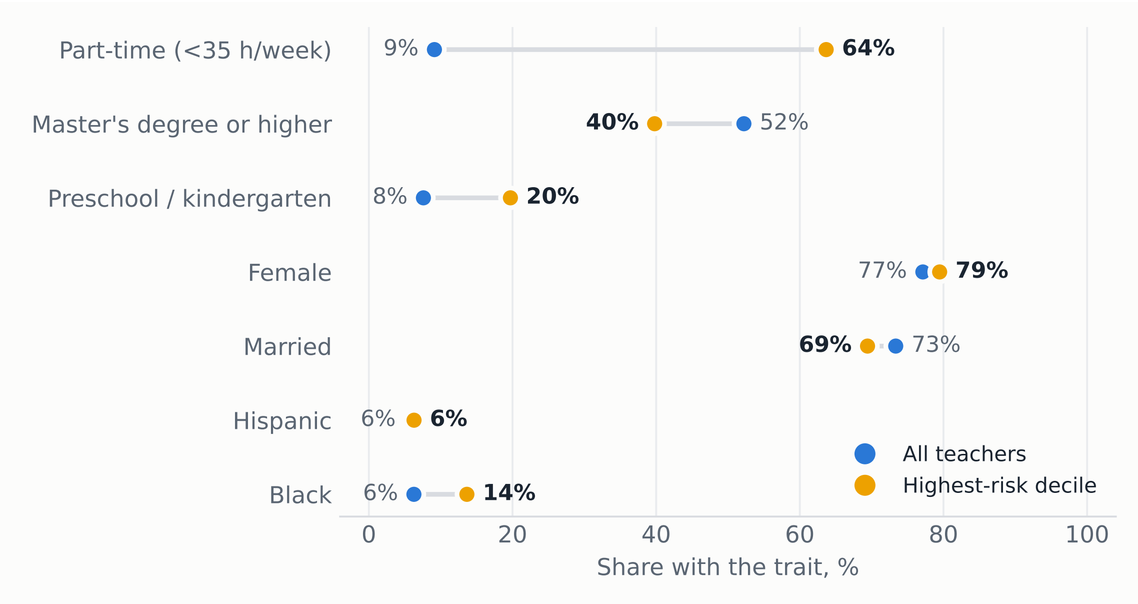
	Probit coef.	(SE)	AME (pp)
Age (years)	-0.062***	(0.004)	-1.36***
Age^2/100	0.079***	(0.004)	+1.73***
Female	-0.039**	(0.017)	-0.86**
Married	0.017	(0.017)	+0.38
Black	0.273***	(0.026)	+5.93***
Hispanic	0.069**	(0.028)	+1.50**
Non-citizen	0.351***	(0.052)	+7.63***
Master's degree or higher	-0.049***	(0.014)	-1.06***
Part-time (<35 h/week)	0.561***	(0.020)	+12.19***
Hours not reported	0.334***	(0.022)	+7.26***
Family income $75k+	-0.116***	(0.017)	-2.53***
Preschool / kindergarten	0.191***	(0.025)	+4.16***
Secondary school	-0.039**	(0.017)	-0.84**
Special education	0.077***	(0.026)	+1.67***
Constant	-0.039	(0.090)	
Base-year fixed effects		Yes	
Observations		129,897	
Pseudo R^2		0.055	

Outcome: persistent leaver. Cluster-robust standard errors (by household) in parentheses. AME: average marginal effect in percentage points. Sample: school teachers with a bachelor's degree or higher and a potential re-interview after $t+12$ (baseline MIS 1–3). Reference categories: full-time, bachelor's degree, elementary/middle school.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

6. The typical leaver

Ranking teachers by predicted risk, the top decile has a mean exit probability of 35%, against 15% for the average teacher. It is overwhelmingly female (80%), late-career (mean age 56), seven times more likely to work part-time (64% vs 9%), less likely to hold a graduate degree (40% vs 52%) and over twice as concentrated in preschool/kindergarten (20% vs 8%), with Black teachers over-represented (14% vs 6%).

Figure 9: Characteristics of the top predicted-risk decile vs all teachers.



Source: predicted probabilities from the probit of Table 2.

7. Policy implications

The results locate attrition in weak attachment (part-time hours, the early-childhood segment, both career extremes) rather than in broad demographics, and show a rising trend since the early 2010s. Four levers follow. First, converting involuntary part-time positions to full-time attacks the single largest risk factor. Second, retention packages for preschool and kindergarten teachers target the most exposed level. Third, the Black, Hispanic and non-citizen retention gaps deserve monitoring, since they compound into the diversity of the future workforce. Fourth, with one in six leavers back within three months, cheap re-entry pathways such as fast-track rehiring, credential portability and return bonuses can recover part of the outflow at low cost.

Methods in one line. Twenty linked CPS 4-8-4 rotation panels, 2005→2025 · 174,872 teachers (129,897 with follow-up) · persistent-leaver outcome · probit, year FE, household-clustered SE · fully reproducible: `src/02_build_cps_panel.py` → `src/05_evolution.py` → `src/03_model_attrition.py`.